

# SQL Case Study

## US Monthly Retail Sales — Analysis & Business Insights

### 1. Overview

This case study is built around real-world-style US monthly retail sales data, spanning multiple years and covering a wide range of industries and business types from automotive dealers to clothing stores. It's designed to give you hands-on practice answering realistic business questions using SQL: aggregations, window functions, CTEs, self-joins, and conditional logic.

The goal isn't just to write correct queries, it's to practice translating a business question into an analytical approach, the way you would on the job as a data analyst or data engineer. Once you're done, this project (your queries, your write-up of findings, and any charts you build on top) is a solid addition to a data analytics portfolio.

### 2. Dataset

**File:**

Dataset Link: [us\\_monthly\\_retail\\_sales\\_wrangled.csv](#)

SQL Analysis Template: [analysis\\_template.sql](#)

*Suggested table name when you load it into your database: retail\_sales*

| Column           | Type    | Description                                               |
|------------------|---------|-----------------------------------------------------------|
| month            | Integer | Month number (1–12)                                       |
| year             | Integer | Calendar year                                             |
| naics_code       | Text    | North American Industry Classification System code        |
| kind_of_business | Text    | Specific business type (e.g. "Women's clothing stores")   |
| industry         | Text    | Broader industry grouping (e.g. "Automotive", "Clothing") |
| sales            | Numeric | Monthly retail sales value (in millions of USD)           |

*Note: sales figures are already cleaned/wrangled for this exercise — you can load the CSV directly into a table named retail\_sales in Postgres, MySQL, SQLite, BigQuery, or any SQL engine of your choice.*

### 3. How to Use This Case Study

- Load the CSV into a SQL environment of your choice (PostgreSQL, MySQL, SQLite, DuckDB, BigQuery, etc.) as a table named retail\_sales.
- Work through the business questions below in order — later questions build on techniques used in earlier ones.
- Try to solve each question yourself before reading the hint. The hint only points you toward the right technique, not the full query.
- Keep your final queries in a single .sql file (a blank template with just the questions and hints is included separately) — this becomes your submission or portfolio artifact.
- Optional stretch goal: turn a few of your query outputs into charts (e.g. a seasonality heatmap, a YoY growth line chart) using a BI tool, Python, or a spreadsheet.

## 4. Business Questions

Answer each question with one SQL query (or a short chain of CTEs). Hints are meant as a nudge, not a solution — try to reason through the approach yourself first.

### Section 1 — Industry Performance & Trends

**Q1. What were the top-performing industries (by total sales) for each month in 2021? How did their sales compare month-over-month?**

*Hint: Group sales by year, month, and industry first (a CTE). Then use `RANK()` OVER (`PARTITION BY` year, month `ORDER BY` total\_sales DESC) in a second step, and filter to rank = 1.*

**Q2. Repeat the same analysis (top industry per month) for the year 2022.**

*Hint: Same logic as Q1 — just change the year filter.*

**Q3. Repeat the same analysis (top industry per month) for the year 2020.**

*Hint: Same logic as Q1 — just change the year filter.*

**Q4. Repeat the same analysis (top industry per month) for the year 2019. Bonus: can you write ONE query that returns the top industry per month across ALL years, instead of one query per year?**

*Hint: Same logic as Q1. For the bonus, remove the `WHERE` year filter entirely and keep year in the `PARTITION BY`.*

### Section 2 — Business Type Contribution

**Q5. Which specific kind of business contributes the most to total sales overall, and how does contribution vary across industries?**

*Hint: A simple `GROUP BY` on kind\_of\_business and industry with `SUM(sales)`, ordered descending — no window functions needed.*

### Section 3 — Seasonality

**Q6. Is there seasonality in sales for specific industries? How does each industry perform month-over-month across years?**

*Hint: Group by year, industry, and month. Once you have the output, think about pivoting month vs. year (in a spreadsheet or BI tool) to visually spot seasonal peaks.*

### Section 4 — NAICS Classification

**Q7. How does the sales distribution vary among industries based on their NAICS codes?**

*Hint: Group by naics\_code and industry, `SUM(sales)`, and order by naics\_code so related codes sit together.*

### Section 5 — Outlier Detection

**Q8. Are there any outliers or sudden spikes/drops in sales for specific industries during particular months or years?**

*Hint: Use `LAG()` and `LEAD()` OVER (`PARTITION BY` industry `ORDER BY` year, month) to compare each month against the previous and next. Define your own outlier threshold (e.g. 1.5x) and justify your choice.*

### Section 6 — Business-Level Averages

**Q9. Which kind of business categories have an all-time average monthly sale above a high threshold (e.g. \$10,000 million)?**

*Hint: GROUP BY kind\_of\_business, use AVG(sales), and filter with HAVING (not WHERE — it's a filter on an aggregate).*

## Section 7 — Automotive Industry Deep-Dive

**Q10. Within the Automotive industry, which kind of business had the highest sales revenue in 2022?**

*Hint: Filter WHERE industry = 'Automotive' AND year = 2022, then GROUP BY kind\_of\_business.*

**Q11. What percentage does each business type contribute to total Automotive industry sales in 2022?**

*Hint: Build one CTE for per-business sales and another for the overall Automotive total, then CROSS JOIN and compute (business\_total / overall\_total) \* 100.*

## Section 8 — Year-over-Year Growth

**Q12. What is the year-over-year (YoY) growth rate for each industry, for every year available?**

*Hint: Try two approaches: (a) self-join a yearly-totals CTE to itself on curr.year = prev.year + 1, matched by industry, or (b) LAG(sales\_sum) OVER (PARTITION BY industry ORDER BY year) in a single pass. Compare which reads more cleanly.*

## Section 9 — Women's vs. Men's Clothing Stores

**Q13. What are the yearly total sales for women's clothing stores and men's clothing stores, side by side?**

*Hint: Use conditional aggregation: SUM(CASE WHEN kind\_of\_business = '...' THEN sales ELSE 0 END), once per store type, grouped by year.*

**Q14. What is the yearly ratio of women's clothing sales to men's clothing sales?**

*Hint: Wrap the previous question's logic in a subquery/CTE, then divide the two computed columns in the outer query.*

**Q15. What is the year-to-date (YTD) cumulative sales, month by month, for women's clothing stores in 2019, 2020, 2021, and 2022?**

*Hint: A correlated subquery (sum sales WHERE year matches AND month <= current row's month) works, but also try rewriting with SUM(sales) OVER (PARTITION BY year ORDER BY month) — a window function will likely be cleaner and faster.*

**Q16. What is the month-over-month growth rate for women's clothing stores in 2022?**

*Hint: LAG(sales, 1) OVER (ORDER BY month) gets the previous month's sales; growth rate = (current - previous) / previous \* 100.*

## 5. Portfolio Tip

When you add this to your portfolio (GitHub, personal site, resume project section), don't just paste the raw SQL. Include:

- A short README explaining the dataset and the business questions you tackled.
- Your final queries, organized by section.
- 2–3 key findings written in plain English (e.g. "Automotive consistently leads monthly sales, but Clothing shows the strongest seasonal spike in December").
- If possible, one or two charts/visuals summarizing a key insight — this is often what recruiters actually look at first.